



α Syn RT-QuIC rapid and ultra-sensitive quantitation of disease-associated α -synuclein seeds in PPMI cerebrospinal fluid

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Summary

The diagnosis and treatment of Parkinson's disease (PD) would benefit from the use of direct and highly sensitive tests to detect the abnormal protein α -synuclein (α Syn). We recently tested our improved α -synuclein real time quaking-induced conversion (α Syn RT-QuIC) assay against a panel of cerebrospinal fluid specimens from patients with clinically typical moderate-to-advanced Parkinson's disease and healthy controls (MJ Fox Foundation/NINDS BioFIND collection). Our assay, which only takes 1-2 days and allows quantitation of misfolded α Syn, showed 97% sensitivity and 87% specificity.

In this study, we used our α Syn RT-QuIC to test the PPMI cerebrospinal fluid (CSF) samples blinded. We performed an initial screening of the samples and identified strong α Syn RT-QuIC positives that were further analyzed by end-point dilution to estimate the seeding dose 50 (SD₅₀: α Syn seeding dose that caused 50% of replicate wells to be positive). We are sending the data back with this report to MJFF/NINDS for decoding.

Method

The PPMI CSF panel was tested blinded by α -synuclein RT-QuIC assay. Samples were thawed at room temperature and immediately used to seed reactions that each contained 85 μ L the following mix (written as the final concentration in 100 μ L): 40 mM phosphate buffer (pH 8.0), 170 mM NaCl, 0.1 mg/ml recombinant K23Q α Syn, 10 μ M thioflavin T (ThT) and 0.0015% sodium dodecyl sulfate (SDS). All the solutions and recombinant K23Q α -Syn substrate batches were tested for adequate performance prior to use.

The testing was done using BMG FLUOstar Omega plate readers with cycles of 1 min shaking (400 rpm double orbital) and 1 min rest throughout the indicated incubation time. Thioflavin T (ThT) fluorescence measurements (450 $\pm$ 10 nm excitation and 480 $\pm$ 10 nm emission; bottom read) were taken every 45 minutes. To account for differences from run to run and between different plate readers, each experimental plate was analyzed separately. A 5-hour baseline, consisting of the window from $\sim$ 2-7 hours was subtracted from each curve. The maximum ThT value on the plate within 48-hours was then used to normalize all curves from 0 (the lowest value for an individual curve) to 100% (the maximum value on the plate). A 10% threshold was set as the criteria for a well to be considered positive. The maximum value for





each well was collected as a percent and was then given a positive/negative determination based on the 10% threshold. For a sample to be considered positive, greater than 25% of the replicate wells needed to be positive. Samples with 25% of the replicate wells positive were considered inconclusive and re-tested. If upon repeat they had 25% or less wells positive they were considered negative and marked as “negative (inconclusive)”. Samples with no positive wells were considered negative. Time to threshold was calculated as the amount of time needed for a well to cross that 10% threshold. Time to half max was calculated by dividing the maximum value for a well by 2 and then finding the time value closest to when that value was reached. Area under the curve, along with each of the other described parameters above (except for time to half max which was calculated manually using Microsoft excel), was calculated using the OMEGA MARS software V3.32 Build 5. The same baseline subtracted, scaled curves were used to calculate area under the curve. All ThT negative wells were reported as “NA”.

End-point dilution analysis of was by α -synuclein RT-QuIC on positive CSFs by serially diluting the sample 2-fold in a negative CSF background. We limited this type of analysis to samples that had all replicate wells positive in the initial screening. We currently have preliminary evidence that it might be worth doing end-point dilution analysis on a subset of CSFs that didn't have all replicates wells positives in the initial screening to better estimate their level of seeding activity in these samples. We would be happy to do that if such information is considered of interest. As of now the SD_{50} per 15uL of CSFs that we identified as positives range from 14.1 to ≥ 95.2 .

References

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